Vibe Code Arena Problem Statements

1. **Smart System for Detecting Phishing and Cyber Fraud -** People regularly receive suspicious messages, emails, website links and calls that attempt to steal personal information, banking details or account credentials. Limited cyber awareness and delayed reporting can allow these scams to cause financial loss or unauthorized access. Develop a cybersecurity tool that allows users to check suspicious emails, SMS messages, website URLs and screenshots for signs of phishing or fraud. The system should examine indicators such as misleading links, fake identities, suspicious domains and requests for confidential information, then provide a risk assessment with a clear explanation of the warning signs. Users should receive practical advice on whether to avoid a link, verify a sender or take further precautions. The platform should also provide educational material on safe browsing, password security and common fraud techniques. A reporting feature should let users record suspected incidents, attach relevant evidence and access appropriate reporting channels. Where supported, users should be able to track the status of submitted reports. An administrative dashboard should summarize reported incidents and recurring patterns while protecting personal information. The solution should be tested for detection accuracy and false alarms, and should handle suspicious links safely without requiring users to open them.
2. **Unified Government Scheme and Housing Assistance Finder -** Citizens, particularly those in rural and underserved communities, may find it difficult to identify government welfare schemes that match their needs, understand eligibility requirements, and access reliable application information. Develop **SchemeSaathi**, a user-friendly portal that helps citizens discover relevant government schemes using basic personal and family details. Users should be able to enter their state, rural or urban location, age group, occupation, approximate income range, and relevant family information. The system should match these details against a predefined database of schemes covering education, healthcare, agriculture, employment, housing, and social welfare, including relevant rural and urban housing assistance programmes such as PMAY. For each recommended scheme, display its purpose, intended beneficiaries, available benefits, eligibility criteria, required documents, and official application links. Classify the results as Potentially Eligible, More Information Required, or Eligibility Criteria Not Met, with clear explanations of how each result was determined. The portal should also generate document checklists and allow users to shortlist schemes for future reference. The prototype should use sample records and transparent, rule-based eligibility checks, with a responsive interface suitable for mobile and desktop devices. It must clearly inform users that the results are preliminary

recommendations and do not guarantee official eligibility, application approval, or receipt of benefits.

3. **Morphed Image Abuse Reporting Assistant -** Women and girls who face manipulated images, online impersonation, or fabricated content may struggle to document the incident, preserve evidence, report the abuse, request content removal, and secure their accounts. Privacy concerns and uncertainty about the reporting process can further discourage victims from seeking help. Develop **NariSuraksha AI**, a privacy-first platform that guides users through the process of documenting incidents and understanding the appropriate next steps. The system should provide a guided incident form to record the type of abuse, affected platform, approximate date, relevant URLs, and incident description. It should include an evidence checklist covering screenshots, links, message records, and timestamps, without requiring sensitive photographs to be uploaded to external servers. Users should be able to organise their records into an incident timeline and access cybercrime reporting guidance, official resources, content-removal checklists, and account-security recommendations. An AI-assisted guidance feature should answer general questions in simple Hindi or English and help users understand available reporting options. The prototype should use fictional reports and simulated workflows, with appropriate safeguards for sensitive information and clear explanations of what data is stored and how it is handled. The system must not claim to verify the authenticity of images or automatically submit official complaints. The expected outcome is a privacy-focused, responsive platform demonstrating guided incident reporting, evidence organisation, timeline creation, AI-assisted guidance, and access to reliable reporting resources.
4. **Cybersecurity Verification and Awareness Tool -** People regularly receive suspicious emails, SMS messages, website links, and calls designed to steal personal information, banking details, or account credentials. Limited cybersecurity awareness and delays in reporting these incidents can lead to financial losses and unauthorized access to accounts. Develop a cybersecurity tool that allows users to check suspicious emails, text messages, website URLs, and uploaded screenshots for signs of phishing or fraud. The system should identify indicators such as suspicious domains, misleading links, impersonation attempts, and requests for confidential information, then provide a risk assessment with clear explanations of the warning signs. It should recommend practical next steps, such as avoiding suspicious links, independently verifying the sender, or securing affected accounts. The platform should also provide educational resources on safe browsing, password security, phishing techniques, and common cyber fraud

methods. Users should be able to record suspected incidents, attach relevant evidence, access appropriate reporting channels, and track report status where supported. An administrative dashboard should summarise reported incidents and recurring scam patterns while restricting access to personal information. The prototype should use sample messages, mock reports, and safe test URLs, and should analyse links as text or through safe reputation checks without automatically opening suspicious websites. It must not collect passwords, one-time passwords (OTPs), or unnecessary personal information. The system should be tested for detection accuracy and false positives, with a clear disclaimer that automated assessments are guidance and cannot guarantee that a message or link is safe or fraudulent. The expected outcome is a functional prototype combining multi-format phishing detection, explainable risk assessments, practical safety recommendations, cybersecurity learning resources, incident reporting, and a basic administrative dashboard.

5. **Smart Counterfeit Product Detection Platform -** Counterfeit products across industries such as cosmetics, medicines, electronics, clothing, and everyday consumer goods can cause financial losses, damage brand reputation, and create serious safety risks. Since counterfeit packaging, labels, barcodes, and QR codes can closely resemble those of genuine products, consumers and businesses may find it difficult to verify authenticity. Develop **TrustMark**, a smart platform that helps users identify potentially counterfeit products by checking product information, analysing packaging details, and verifying available authenticity records. Users should be able to scan a QR code or barcode, or enter the product identifier manually, to check its details against a sample database of registered products. The system should flag unknown or invalid identifiers and detect codes that appear repeatedly in suspicious circumstances. It should also allow users to upload or capture product and packaging images for comparison with reference images, examining visible details such as logos, labels, spelling, and packaging differences where feasible. Based on the available checks, the platform should generate an explainable authenticity risk assessment and classify results as Likely Genuine, Needs Further Verification, or Potentially Counterfeit, with clear explanations of any warnings. The database should maintain product records, including brand, product ID, batch number, and reference images, with authorised administrators able to add or update entries. Consumers should be able to report suspected counterfeit products by providing images and basic purchase details, while an administrative dashboard supports report review and tracking. The platform should also maintain verification history and display previous warnings associated with suspicious product identifiers. The prototype should use

sample product records and simulated reports, with a responsive interface and a suitable technology stack such as React or HTML, CSS, and JavaScript for the frontend, Node.js with Express for the backend, and Firebase, Supabase, or MongoDB for data storage. QR-code scanning and optional image analysis using OpenCV or a pre-trained model may be included. The system must clearly communicate that barcodes and QR codes can be copied, and that image comparisons and automated risk scores cannot provide definitive proof of authenticity. The expected outcome is a functional prototype that verifies product identifiers, flags suspicious codes, provides explainable risk assessments, supports optional packaging-image analysis, and enables consumers and administrators to report and review suspected counterfeit products.

6. **AI-Powered Smart Queue Management Platform -** Long waiting times at hospitals, banks, government offices, and service centres can cause inconvenience, overcrowding, and inefficient use of time. Traditional queue systems often provide limited real-time information, making it difficult for people to estimate their waiting time, manage appointments, and plan their visits. Develop **QueueSmart AI**, an intelligent queue management platform that helps organisations manage customer queues, estimate waiting times, and monitor service activity through a centralised dashboard. Users should be able to join a queue by selecting a service category and entering basic details, after which the system generates a unique token number. Waiting-time estimates should be calculated using the number of people ahead, average service duration, and available service counters. A live queue dashboard should display the currently serving token, waiting tokens, completed services, and estimated waiting times. The platform should analyse queue length and service patterns to suggest suitable counter allocation and identify potential overcrowding. It should also support appointment slots and configurable priority categories, with clear, predefined rules for assigning priority. Simulated notifications should inform users when their turn is approaching or when their estimated waiting time changes. An administrative dashboard should allow authorised staff to manage queues, add or close service counters, call the next token, and view basic operational statistics. The prototype should use sample customer records and simulated queue data, with a responsive interface suitable for users and service staff. Waiting-time predictions must be presented as estimates rather than guarantees, and priority handling should follow transparent rules. Real SMS integrations, payment systems, and advanced predictive models are not required. The expected outcome is a functional prototype demonstrating digital queue registration, token generation,

waiting-time estimation, live queue monitoring, basic queue optimisation, simulated notifications, and administrative queue management.